

# HR Analytics: Predicting Employee Attrition

Exploratory analysis and machine learning on the IBM HR Analytics dataset to find why employees leave and to predict who is likely to resign.

 Python  scikit-learn  SHAP  Power BI

## Project Overview

Employee attrition is costly because hiring and training replacements takes time and money. This project analyses **1,470 employee records** to:

1. Show how attrition varies by department, overtime, income and age.
2. Train a model that predicts whether an employee will leave.
3. Explain which factors drive the predictions, using SHAP.

## Key Results

| Metric                 | Value               |
|------------------------|---------------------|
| Employees analysed     | 1,470               |
| Overall attrition rate | 16%                 |
| Model                  | Logistic Regression |
| Test accuracy          | 86%                 |
| Recall for leavers     | 0.34                |
| Precision for leavers  | 0.62                |

**Honest takeaway:** the data is imbalanced (only 16% leavers), so accuracy alone looks better than the model really is. The model finds about 1 in 3 real leavers. Balancing the classes is the main next step.

## Key Insights

- **Overtime is the strongest signal.** 30.5% of employees working overtime leave, compared with 10.4% of those who do not.
- **Department matters.** Attrition is highest in Sales (20.6%) and Human Resources (19.0%), and lowest in Research & Development (13.8%).
- **Pay and age.** Leavers have lower monthly income (median about 3,200 vs about 5,200) and are

mostly in their 20s and 30s.

- **SHAP drivers.** Overtime, frequent business travel, fewer total working years and long gaps since the last promotion push predictions towards leaving.

## Charts

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## Methodology

1. **Load and check** the data with pandas (missing values, data types).
2. **Explore** attrition by department, overtime, income and age.
3. **Prepare** the data: one-hot encode categorical columns and scale numeric features.
4. **Split** into 1,176 training and 294 test records (80/20).
5. **Train** a Logistic Regression model (`max_iter=1000`).
6. **Evaluate** with accuracy, precision, recall, F1-score and a confusion matrix.
7. **Explain** the model with SHAP values.
8. **Export** a cleaned file for a Power BI dashboard.

## Repository Contents

| File                                               | Description                                |
|----------------------------------------------------|--------------------------------------------|
| <code>attrition_analysis.ipynb</code>              | Full analysis and model (Jupyter Notebook) |
| <code>WA_Fn-UseC_-HR-Employee-Attrition.csv</code> | Original IBM HR Analytics                  |

| File                                                               | Description                                  |
|--------------------------------------------------------------------|----------------------------------------------|
|                                                                    | dataset                                      |
| <code>hr_data_for_powerbi.csv</code>                               | Cleaned data used for the Power BI dashboard |
| <code>chart1_department.png</code> to <code>chart6_shap.png</code> | Charts used in the report                    |
| <code>HR_Attrition_Report.pdf</code>                               | Full project report                          |

## How to Run

```
# 1. Clone the repository
git clone https://github.com/DILSHA-A/hr-attrition-project.git
cd hr-attrition-project

# 2. Install the libraries
pip install pandas numpy matplotlib seaborn
scikit-learn shap jupyter

# 3. Open the notebook
jupyter notebook attrition_analysis.ipynb
```

## Limitations and Next Steps

- Improve recall for leavers with SMOTE, class weights or oversampling.
- Tune the decision threshold to trade some precision for higher recall.
- Compare Random Forest and Gradient Boosting using cross-validation.
- The dataset is a single snapshot, so results show association, not cause.

## Recommendations for HR

- Review workloads for employees who regularly work overtime.
- Check pay for lower-income and early-career staff.
- Build clear promotion paths.
- Focus retention efforts on Sales, HR and younger employees.

## Author

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## Dataset Source

IBM HR Analytics Employee Attrition & Performance dataset (publicly available, fictional data created by

IBM data scientists).